Module 3: Forward Propagation, Backpropagation and Optimization

1. The Complete Training Process

A neural network learns by repeatedly making predictions, measuring its errors, calculating gradients, and updating its trainable parameters.

The main concepts are:

1. Forward propagation
2. Loss function
3. Backpropagation
4. Optimizer
5. Repeated training over batches and epochs

2. Forward Propagation

Forward propagation is the process of passing input data through the neural network to calculate its output.

The data moves from the input layer through the hidden layers and finally to the output layer.

Input Data

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Input Layer

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Hidden Layer 1

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Hidden Layer 2

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v

Output Layer

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Prediction

During forward propagation, the network uses its current weights and biases to calculate the prediction.

The parameters are not updated during this step.

3. Mathematical Representation

For a dense neural network layer:

z = Wx + b

The activation is then calculated:

a = f(z)

Where:

- W = weight matrix
- x = input vector
- b = bias vector
- z = weighted sum before activation
- f = activation function
- a = layer output or activation

The output of one layer becomes the input to the next layer.

4. What Is a Loss Function?

A loss function measures how different the model's prediction is from the target according to a chosen mathematical objective.

For example, suppose the actual value is 1 and the model predicts 0.2.

The prediction is not close to the target, so an appropriate loss function will assign it a penalty.

The exact penalty depends on the selected loss function.

Common Loss Functions

Mean Squared Error (MSE)

Often used for regression.

$MSE = (1/n) \times \Sigma(y_i - \hat{y}_i)^2$

Here:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of examples

Cross-Entropy Loss

Commonly used for classification tasks.

It penalizes predictions that assign low probability to the correct class.

Important Difference

Loss and accuracy are not the same.

- Loss measures prediction error according to a mathematical function.
- Accuracy measures the fraction of predictions that are classified correctly.

A model's loss can improve even when its accuracy does not change immediately.

5. What Is Backpropagation?

Backpropagation is an algorithm that calculates gradients of the loss with respect to the trainable parameters of a neural network.

It uses the chain rule from calculus to determine how changes in weights and biases affect the loss.

In simple terms, it helps determine which parameter changes could reduce the error.

Input

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Forward Propagation

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Prediction

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v

Calculate Loss

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Backpropagation

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v

Calculate Gradients

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Optimizer Updates Parameters

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Next Training Step

6. What Is the Chain Rule?

The chain rule is a calculus rule used to calculate the derivative of a function that depends on another function.

Neural networks contain multiple layers of calculations. The chain rule allows gradients to be propagated backward through these layers.

For example, if:

$y = f(u)$

and

$u = g(x)$

then:

$$\mathbf{\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}}$$

Backpropagation applies this principle repeatedly to calculate gradients for the parameters of the network.

7. What Is an Optimizer?

An optimizer uses the calculated gradients to update the model's trainable parameters.

Common optimizers include:

- Gradient Descent
- Stochastic Gradient Descent (SGD)
- SGD with Momentum
- Adam

Gradient Descent Update Rule

A basic gradient descent update is:

New Parameter = Old Parameter − Learning Rate × Gradient

Or:

$$\theta_{new} = \theta_{old} - \eta \times \left(\frac{\partial L}{\partial \theta}\right)$$

Where:

- θ = a trainable parameter, such as a weight
- η = learning rate
- L = loss function
- $\frac{\partial L}{\partial \theta}$ = gradient of loss with respect to that parameter

The optimizer uses the gradient to decide the direction and size of the update according to its update rule.

8. What Is the Learning Rate?

The learning rate controls the step size of parameter updates.

If the Learning Rate Is Too High

- Updates may overshoot useful parameter values.
- Training may become unstable.
- The loss may fluctuate or increase.

If the Learning Rate Is Too Low

- Training may progress very slowly.
- More training steps may be required.

Choosing a suitable learning rate is an important part of training a neural network.

9. What Is an Epoch?

An epoch represents one complete pass through the training dataset.

For example, suppose a training dataset contains 1,000 examples.

If the model processes all 1,000 examples once, it has completed one epoch.

Training for multiple epochs allows the model to make repeated parameter updates using the training data.

10. What Is a Batch?

A batch is a subset of training examples processed together during a training step.

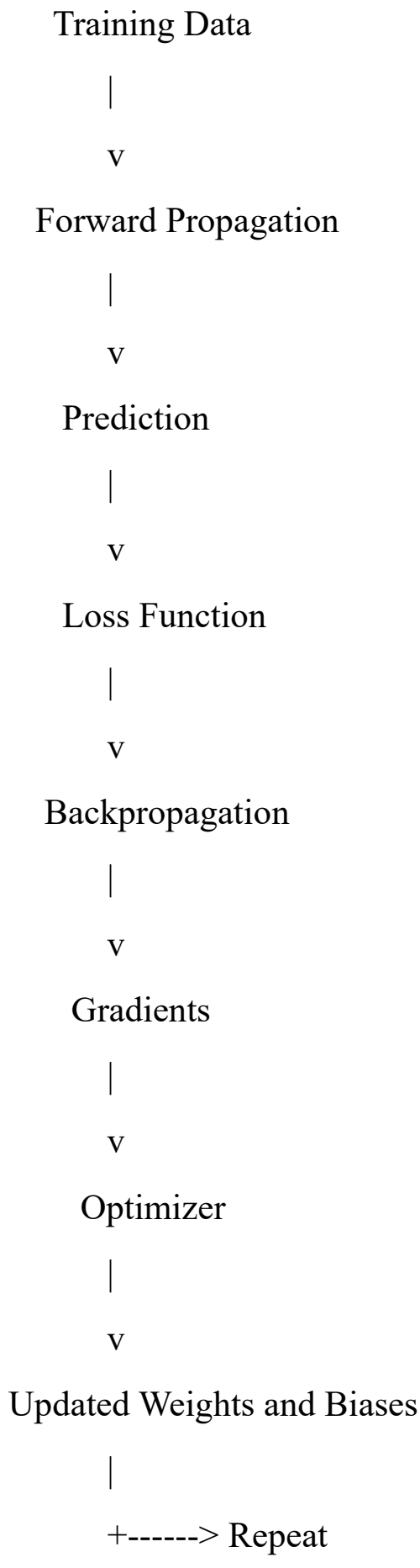
If there are 1,000 training examples and the batch size is 100, one epoch typically contains 10 batches.

After each batch, the model can calculate gradients and update its parameters.

11. Complete Training Workflow

The training process can be summarized as follows:

1. Initialize weights and biases.
2. Take a batch of training examples.
3. Perform forward propagation.
4. Calculate the loss.
5. Perform backpropagation to calculate gradients.
6. Use the optimizer to update weights and biases.
7. Repeat for the remaining batches.
8. Repeat for the required number of epochs.
9. Evaluate performance on validation data that is kept separate from training.


Training Data
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Forward Propagation
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Prediction
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Loss Function
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v
Backpropagation
|
v
Gradients
|
v
Optimizer
|
v
Updated Weights and Biases
|
+------> Repeat

12. Important Interview Clarification

Forward propagation, loss calculation, backpropagation, and optimization are related but different operations.

- **Forward propagation:** Calculates predictions.
- **Loss function:** Measures prediction error.
- **Backpropagation:** Calculates gradients.
- **Optimizer:** Uses gradients to update parameters.

Backpropagation does not directly mean that weights are updated. The optimizer performs the parameter update using the gradients.

13. Quick Revision Questions

1. What happens during forward propagation?
2. What is a loss function?
3. What does backpropagation calculate?
4. What is the chain rule?
5. What is the difference between backpropagation and an optimizer?
6. What is the learning rate?
7. What is the difference between a batch and an epoch?

Module Summary

A neural network performs forward propagation to make predictions. A loss function measures the error. Backpropagation calculates gradients using the chain rule, and an optimizer uses these gradients to update weights and biases. Repeating this process helps the network learn from its training data.